

回溯法

回溯法理解：

最简单的方法就是把它看作是用递归实现的多重循环。

理解前提：对递归的理解很重要。

每个子程序尽管名字相中，但它们均在不同的空间中运行，互不相关。

每个子程序内部声明的容器为其私有。自动随子程序的调用而创建，随子程序的返回而销毁。

主程序会暂停，等待子程序运行结束，再继续运行。

子程序调用在程序中只表现为一个调用。但它实际运行过程非常复杂，实质是一棵树的遍历。

递归回溯通用框架(背住)

procedure dfs(i:longint)第i步(第i重循环);

var

x:longint; //思考：为什么x只是能局部变量？(第i重循环的循环变量)

begin

if i=maxi then 输出并返回 //递归边界(深度控制、循环层数控制)

for x:=1 to max //当前第i步共有max种走法，每一种都要试一试。

begin

if 没有冲突 then //判断当前第i步，第x种走法是否可行

begin

记录方案

宣布占领

dfs(i+1)//看起来只是一个语句，实质是一棵递归子树。

宣布释放//（回溯，此处是回溯法的关键。当前第i步需撤销第x种走法，以便第i步试第x+1种走法）

end;

end;

end;

一句话理解回溯法：递归回溯法=多重循环

例题一：生成N的全排列（由1到N组成的N个数的不同序列，每个数只能用一次）

输入样例：

3

输出样例：

1 2 3

1 3 2

2 1 3

2 3 1

3 1 2

3 2 1

分析：N的全排列问题可以这样来思考，可以看作是分为N步来完成，每步确定一个数。第i个数可以是1到N，但不能与前边的其它数冲突。可以用N重循环来实现。

算法：回溯法

program quanpailie;

const

maxn=100;

var

n:longint;

a:array [1..maxn]of longint;//a[i]:第i个数

used:array [1..maxn]of boolean;//used[i]:数字i是否被使用过

procedure writeans;

var i:longint;

begin

for i:=1 to n do write(a[i], ' ');

writeln;

end;

procedure dfs(i:longint);//确定第i个数

var

x:longint;

begin

```

if i>n then//递归边界
begin
    writeans;
    exit;
end;

for x:=1 to n do//第i个数可以是1..n
begin
    if used[x]=false then//可行性判断：第i个数若为x，前提是它必须没有被使用过
    begin
        a[i]:=x;//记录方案，第i个数为x
        used[x]:=true;//宣布x被使用了
        dfs(i+1);//确定下一个数(它不是简单的一句话，它是一棵子树，会把后边所有的数放完)
        used[x]:=false;//回溯，第i个数要用下一个数，不用x了
    end;
end;
end;

begin{主程序}
    Readln(n);
    fillchar(used, sizeof(used), 0);
    dfs(1);
end.

```

例2、跳马问题。在5*5格的棋盘上，有一个国家象棋的马，它可以朝8个方向跳，但不允许出界或跳到已跳过的格子上，要求不重复地跳遍整个棋盘。

```

program jump;
//用h[i, j]表示地图上坐标为i, j的位置。
//h[i, j]=0: 没跳过;    h[i, j]=i: 第i步在这个位置
var
    h:array[-1..7, -1..7] of longint;
    a, b:array[1..8] of longint;//a[i], b[i]: 马第i种跳法x, y坐标的增量
    i, j, num:longint;//num:方法总数

procedure print;//输出
var
    i, j:longint;
begin
    num:=num+1;//累加方案总数
    if num<=5 then//为了调试方便，该程序只输出了前5种方案
    begin
        for i:=1 to 5 do
        begin
            for j:=1 to 5 do write(h[i, j]:4);
            writeln;
        end;
        writeln;
    end;
end;

procedure dfs(x, y, i:longint);//从x, y位置跳第i步
var
    j, u, v:longint;
begin
    if i>25 then//递归边界
    begin
        print;
        exit;
    end;

    for j:=1 to 8 do//第i重循环从x, y位置出发，有8个方向，用j来枚举
    begin
        u:=x+a[j];    //按第J种跳法，会跳到u, v位置
        v:=y+b[j];
        if h[u, v]=0 then//可行：位置(u, v)没走过
        begin
            h[u, v]:=i;//第i步跳到u, v
            dfs(u, v, i+1);//走下一步，它是一棵树，会把后边所有方案走完才返回
        end;
    end;
end;

```

```

        h[u,v]:=0//回溯，释放第i步，第i步重新跳下一种走法。
    end;
end;
end;
begin
    {地图加上了一个边框，以免马跳出边界}
    for i:=-1 to 7 do
        for j:=-1 to 7 do
            if (i>=1)and(i<=5)and(j>=1)and(j<=5) then h[i,j]:=0 else h[i,j]:=1;
            {8种跳法，x,y方向上的变化}
            a[1]:=2;b[1]:=1;
            a[2]:=1;b[2]:=2;
            a[3]:=-1;b[3]:=2;
            a[4]:=-2;b[4]:=1;
            a[5]:=-2;b[5]:=-1;
            a[6]:=-1;b[6]:=-2;
            a[7]:=1;b[7]:=-2;
            a[8]:=2;b[8]:=-1;

            num:=0;
            h[1,1]:=1;//从1，1位置开始跳
            dfs(1,1,2);
            writeln('num=',num);
        end.

```

练习(选作)：走迷宫(migong.pas/migong.in/migong.out)
 在一个7*12的迷宫中，编程帮老鼠从左上角入口走到右下角出口
 要求：

- 1:打印所有方案
- 2:输出方案总数
- 3:找出最短路径长度

输入样例：

```

0 0 0 0 0 0 0 1 1 0 1 1
0 0 1 1 1 1 0 0 0 0 0 1
0 0 0 0 0 0 1 1 1 0 1 1
0 0 1 0 1 0 0 0 0 0 1 1
0 0 1 0 0 1 1 0 1 0 1 1
0 0 0 0 0 0 0 0 1 0 0 1
0 1 1 1 1 1 1 1 1 1 0 0

```

其中1代表障碍，0代表无障碍

输出样例：

```

1 0 0 0 0 0 0 1 1 0 1 1
2 0 1 1 1 1 0 0 0 0 0 1
3 0 0 0 0 0 1 1 1 0 1 1
4 0 1 0 1 0 0 15 16 17 1 1
5 0 1 0 0 1 1 14 1 18 1 1
6 7 8 9 10 11 12 13 1 19 20 1
0 1 1 1 1 1 1 1 1 1 21 22

```

```

1 24 25 26 27 28 29 1 1 0 1 1
2 23 1 1 1 1 30 31 32 33 0 1
3 22 21 20 19 18 1 1 1 34 1 1
4 0 1 0 1 17 16 15 0 35 1 1
5 0 1 0 0 1 1 14 1 36 1 1
6 7 8 9 10 11 12 13 1 37 38 1
0 1 1 1 1 1 1 1 1 1 39 40

```

```

1 0 0 0 0 0 0 1 1 0 1 1
2 0 1 1 1 1 0 0 0 0 0 1
3 0 0 14 15 16 1 1 1 0 1 1
4 0 1 13 1 17 18 19 20 21 1 1
5 0 1 12 11 1 1 0 1 22 1 1
6 7 8 9 10 0 0 0 1 23 24 1
0 1 1 1 1 1 1 1 1 1 25 26

```

```

1 18 19 20 21 22 23 1 1 0 1 1
2 17 1 1 1 1 24 25 26 27 0 1
3 16 15 14 0 0 1 1 1 28 1 1

```

```

4 0 1 13 1 0 0 0 0 29 1 1
5 0 1 12 11 1 1 0 1 30 1 1
6 7 8 9 10 0 0 0 1 31 32 1
0 1 1 1 1 1 1 1 1 1 33 34

```

```

1 0 0 0 0 0 0 1 1 0 1 1
2 0 1 1 1 1 0 0 0 0 0 1
3 0 0 0 0 0 1 1 1 0 1 1
4 0 1 0 1 0 0 17 18 19 1 1
5 0 1 10 11 1 1 16 1 20 1 1
6 7 8 9 12 13 14 15 1 21 22 1
0 1 1 1 1 1 1 1 1 1 23 24

```

```

1 26 27 28 29 30 31 1 1 0 1 1
2 25 1 1 1 1 32 33 34 35 0 1
3 24 23 22 21 20 1 1 1 36 1 1
4 0 1 0 1 19 18 17 0 37 1 1
5 0 1 10 11 1 1 16 1 38 1 1
6 7 8 9 12 13 14 15 1 39 40 1
0 1 1 1 1 1 1 1 1 1 41 42

```

```

1 0 0 0 0 0 0 1 1 0 1 1
2 0 1 1 1 1 0 0 0 0 0 1
3 0 0 12 13 14 1 1 1 0 1 1
4 0 1 11 1 15 16 17 18 19 1 1
5 0 1 10 0 1 1 0 1 20 1 1
6 7 8 9 0 0 0 0 1 21 22 1
0 1 1 1 1 1 1 1 1 1 23 24

```

```

1 16 17 18 19 20 21 1 1 0 1 1
2 15 1 1 1 1 22 23 24 25 0 1
3 14 13 12 0 0 1 1 1 26 1 1
4 0 1 11 1 0 0 0 0 27 1 1
5 0 1 10 0 1 1 0 1 28 1 1
6 7 8 9 0 0 0 0 1 29 30 1
0 1 1 1 1 1 1 1 1 1 31 32

```

```

1 0 0 0 0 0 0 1 1 0 1 1
2 0 1 1 1 1 0 0 0 0 0 1
3 10 11 12 0 0 1 1 1 0 1 1
4 9 1 13 1 0 0 21 22 23 1 1
5 8 1 14 0 1 1 20 1 24 1 1
6 7 0 15 16 17 18 19 1 25 26 1
0 1 1 1 1 1 1 1 1 1 27 28

```

```

1 0 0 0 0 0 0 1 1 0 1 1
2 0 1 1 1 1 0 0 0 0 0 1
3 10 11 12 0 0 1 1 1 0 1 1
4 9 1 13 1 0 0 21 22 23 1 1
5 8 1 14 15 1 1 20 1 24 1 1
6 7 0 0 16 17 18 19 1 25 26 1
0 1 1 1 1 1 1 1 1 1 27 28

```

```

1 0 0 0 0 0 0 1 1 0 1 1
2 0 1 1 1 1 0 0 0 0 0 1
3 10 11 12 13 14 1 1 1 0 1 1
4 9 1 0 1 15 16 17 18 19 1 1
5 8 1 0 0 1 1 0 1 20 1 1
6 7 0 0 0 0 0 0 1 21 22 1
0 1 1 1 1 1 1 1 1 1 23 24

```

```

1 12 13 14 15 16 17 1 1 0 1 1
2 11 1 1 1 1 18 19 20 21 0 1
3 10 0 0 0 0 1 1 1 22 1 1
4 9 1 0 1 0 0 0 0 23 1 1
5 8 1 0 0 1 1 0 1 24 1 1
6 7 0 0 0 0 0 0 1 25 26 1
0 1 1 1 1 1 1 1 1 1 27 28

```

```

1 0 0 0 0 0 0 1 1 0 1 1
2 0 1 1 1 1 0 0 0 0 0 1
3 0 0 0 0 0 1 1 1 0 1 1
4 0 1 0 1 0 0 15 16 17 1 1
5 6 1 0 0 1 1 14 1 18 1 1
0 7 8 9 10 11 12 13 1 19 20 1
0 1 1 1 1 1 1 1 1 1 21 22

```

```

1 24 25 26 27 28 29 1 1 0 1 1
2 23 1 1 1 1 30 31 32 33 0 1
3 22 21 20 19 18 1 1 1 34 1 1
4 0 1 0 1 17 16 15 0 35 1 1
5 6 1 0 0 1 1 14 1 36 1 1
0 7 8 9 10 11 12 13 1 37 38 1
0 1 1 1 1 1 1 1 1 1 39 40

```

```

1 0 0 0 0 0 0 1 1 0 1 1
2 0 1 1 1 1 0 0 0 0 0 1
3 0 0 14 15 16 1 1 1 0 1 1
4 0 1 13 1 17 18 19 20 21 1 1
5 6 1 12 11 1 1 0 1 22 1 1
0 7 8 9 10 0 0 0 1 23 24 1
0 1 1 1 1 1 1 1 1 1 25 26

```

```

1 18 19 20 21 22 23 1 1 0 1 1
2 17 1 1 1 1 24 25 26 27 0 1
3 16 15 14 0 0 1 1 1 28 1 1
4 0 1 13 1 0 0 0 0 29 1 1
5 6 1 12 11 1 1 0 1 30 1 1
0 7 8 9 10 0 0 0 1 31 32 1
0 1 1 1 1 1 1 1 1 1 33 34

```

```

1 0 0 0 0 0 0 1 1 0 1 1
2 0 1 1 1 1 0 0 0 0 0 1
3 0 0 0 0 0 1 1 1 0 1 1
4 0 1 0 1 0 0 17 18 19 1 1
5 6 1 10 11 1 1 16 1 20 1 1
0 7 8 9 12 13 14 15 1 21 22 1
0 1 1 1 1 1 1 1 1 1 23 24

```

```

1 26 27 28 29 30 31 1 1 0 1 1
2 25 1 1 1 1 32 33 34 35 0 1
3 24 23 22 21 20 1 1 1 36 1 1
4 0 1 0 1 19 18 17 0 37 1 1
5 6 1 10 11 1 1 16 1 38 1 1
0 7 8 9 12 13 14 15 1 39 40 1
0 1 1 1 1 1 1 1 1 1 41 42

```

```

1 0 0 0 0 0 0 1 1 0 1 1
2 0 1 1 1 1 0 0 0 0 0 1
3 0 0 12 13 14 1 1 1 0 1 1
4 0 1 11 1 15 16 17 18 19 1 1
5 6 1 10 0 1 1 0 1 20 1 1
0 7 8 9 0 0 0 0 1 21 22 1
0 1 1 1 1 1 1 1 1 1 23 24

```

```

1 16 17 18 19 20 21 1 1 0 1 1
2 15 1 1 1 1 22 23 24 25 0 1
3 14 13 12 0 0 1 1 1 26 1 1
4 0 1 11 1 0 0 0 0 27 1 1
5 6 1 10 0 1 1 0 1 28 1 1
0 7 8 9 0 0 0 0 1 29 30 1
0 1 1 1 1 1 1 1 1 1 31 32

```

```

1 0 0 0 0 0 0 1 1 0 1 1
2 0 1 1 1 1 0 0 0 0 0 1
3 8 9 10 0 0 1 1 1 0 1 1
4 7 1 11 1 0 0 19 20 21 1 1
5 6 1 12 0 1 1 18 1 22 1 1

```

```

0 0 0 13 14 15 16 17 1 23 24 1
0 1 1 1 1 1 1 1 1 1 1 25 26

1 0 0 0 0 0 0 1 1 0 1 1
2 0 1 1 1 1 0 0 0 0 0 1
3 8 9 10 0 0 1 1 1 0 1 1
4 7 1 11 1 0 0 19 20 21 1 1
5 6 1 12 13 1 1 18 1 22 1 1
0 0 0 0 14 15 16 17 1 23 24 1
0 1 1 1 1 1 1 1 1 1 1 25 26

1 0 0 0 0 0 0 1 1 0 1 1
2 0 1 1 1 1 0 0 0 0 0 1
3 8 9 10 11 12 1 1 1 0 1 1
4 7 1 0 1 13 14 15 16 17 1 1
5 6 1 0 0 1 1 0 1 18 1 1
0 0 0 0 0 0 0 0 1 19 20 1
0 1 1 1 1 1 1 1 1 1 1 21 22

1 10 11 12 13 14 15 1 1 0 1 1
2 9 1 1 1 1 16 17 18 19 0 1
3 8 0 0 0 0 1 1 1 20 1 1
4 7 1 0 1 0 0 0 0 21 1 1
5 6 1 0 0 1 1 0 1 22 1 1
0 0 0 0 0 0 0 0 1 23 24 1
0 1 1 1 1 1 1 1 1 1 1 25 26

1 0 0 0 0 0 0 1 1 0 1 1
2 0 1 1 1 1 0 0 0 0 0 1
3 0 0 0 0 0 1 1 1 0 1 1
4 5 1 0 1 0 0 15 16 17 1 1
0 6 1 0 0 1 1 14 1 18 1 1
0 7 8 9 10 11 12 13 1 19 20 1
0 1 1 1 1 1 1 1 1 1 1 21 22

1 24 25 26 27 28 29 1 1 0 1 1
2 23 1 1 1 1 30 31 32 33 0 1
3 22 21 20 19 18 1 1 1 34 1 1
4 5 1 0 1 17 16 15 0 35 1 1
0 6 1 0 0 1 1 14 1 36 1 1
0 7 8 9 10 11 12 13 1 37 38 1
0 1 1 1 1 1 1 1 1 1 1 39 40

1 0 0 0 0 0 0 1 1 0 1 1
2 0 1 1 1 1 0 0 0 0 0 1
3 0 0 14 15 16 1 1 1 0 1 1
4 5 1 13 1 17 18 19 20 21 1 1
0 6 1 12 11 1 1 0 1 22 1 1
0 7 8 9 10 0 0 0 1 23 24 1
0 1 1 1 1 1 1 1 1 1 1 25 26

1 18 19 20 21 22 23 1 1 0 1 1
2 17 1 1 1 1 24 25 26 27 0 1
3 16 15 14 0 0 1 1 1 28 1 1
4 5 1 13 1 0 0 0 0 29 1 1
0 6 1 12 11 1 1 0 1 30 1 1
0 7 8 9 10 0 0 0 1 31 32 1
0 1 1 1 1 1 1 1 1 1 1 33 34

1 0 0 0 0 0 0 1 1 0 1 1
2 0 1 1 1 1 0 0 0 0 0 1
3 0 0 0 0 0 1 1 1 0 1 1
4 5 1 0 1 0 0 17 18 19 1 1
0 6 1 10 11 1 1 16 1 20 1 1
0 7 8 9 12 13 14 15 1 21 22 1
0 1 1 1 1 1 1 1 1 1 1 23 24

1 26 27 28 29 30 31 1 1 0 1 1
2 25 1 1 1 1 32 33 34 35 0 1

```

```

3 24 23 22 21 20 1 1 1 36 1 1
4 5 1 0 1 19 18 17 0 37 1 1
0 6 1 10 11 1 1 16 1 38 1 1
0 7 8 9 12 13 14 15 1 39 40 1
0 1 1 1 1 1 1 1 1 1 41 42

```

```

1 0 0 0 0 0 0 1 1 0 1 1
2 0 1 1 1 1 0 0 0 0 0 1
3 0 0 12 13 14 1 1 1 0 1 1
4 5 1 11 1 15 16 17 18 19 1 1
0 6 1 10 0 1 1 0 1 20 1 1
0 7 8 9 0 0 0 0 1 21 22 1
0 1 1 1 1 1 1 1 1 1 23 24

```

```

1 16 17 18 19 20 21 1 1 0 1 1
2 15 1 1 1 1 22 23 24 25 0 1
3 14 13 12 0 0 1 1 1 26 1 1
4 5 1 11 1 0 0 0 0 27 1 1
0 6 1 10 0 1 1 0 1 28 1 1
0 7 8 9 0 0 0 0 1 29 30 1
0 1 1 1 1 1 1 1 1 1 31 32

```

```

1 0 0 0 0 0 0 1 1 0 1 1
2 0 1 1 1 1 0 0 0 0 0 1
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4 5 1 0 1 0 0 17 18 19 1 1
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```

```

1 26 27 28 29 30 31 1 1 0 1 1
2 25 1 1 1 1 32 33 34 35 0 1
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0 1 1 1 1 1 1 1 1 1 41 42

```

```

1 0 0 0 0 0 0 1 1 0 1 1
2 0 1 1 1 1 0 0 0 0 0 1
3 0 0 16 17 18 1 1 1 0 1 1
4 5 1 15 1 19 20 21 22 23 1 1
7 6 1 14 13 1 1 0 1 24 1 1
8 9 10 11 12 0 0 0 1 25 26 1
0 1 1 1 1 1 1 1 1 1 27 28

```

```

1 20 21 22 23 24 25 1 1 0 1 1
2 19 1 1 1 1 26 27 28 29 0 1
3 18 17 16 0 0 1 1 1 30 1 1
4 5 1 15 1 0 0 0 0 31 1 1
7 6 1 14 13 1 1 0 1 32 1 1
8 9 10 11 12 0 0 0 1 33 34 1
0 1 1 1 1 1 1 1 1 1 35 36

```

```

1 0 0 0 0 0 0 1 1 0 1 1
2 0 1 1 1 1 0 0 0 0 0 1
3 0 0 0 0 0 0 1 1 1 0 1 1
4 5 1 0 1 0 0 19 20 21 1 1
7 6 1 12 13 1 1 18 1 22 1 1
8 9 10 11 14 15 16 17 1 23 24 1
0 1 1 1 1 1 1 1 1 1 25 26

```

```

1 28 29 30 31 32 33 1 1 0 1 1
2 27 1 1 1 1 34 35 36 37 0 1
3 26 25 24 23 22 1 1 1 38 1 1
4 5 1 0 1 21 20 19 0 39 1 1
7 6 1 12 13 1 1 18 1 40 1 1
8 9 10 11 14 15 16 17 1 41 42 1
0 1 1 1 1 1 1 1 1 1 43 44

```

```

1 0 0 0 0 0 0 1 1 0 1 1
2 0 1 1 1 1 0 0 0 0 0 1
3 0 0 14 15 16 1 1 1 0 1 1
4 5 1 13 1 17 18 19 20 21 1 1
7 6 1 12 0 1 1 0 1 22 1 1
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```

```

1 18 19 20 21 22 23 1 1 0 1 1
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4 5 1 13 1 0 0 0 0 29 1 1
7 6 1 12 0 1 1 0 1 30 1 1
8 9 10 11 0 0 0 0 1 31 32 1
0 1 1 1 1 1 1 1 1 1 33 34

```

```

1 0 0 0 0 0 0 1 1 0 1 1
2 0 1 1 1 1 0 0 0 0 0 1
3 6 7 8 0 0 1 1 1 0 1 1
4 5 1 9 1 0 0 17 18 19 1 1
0 0 1 10 0 1 1 16 1 20 1 1
0 0 0 11 12 13 14 15 1 21 22 1
0 1 1 1 1 1 1 1 1 1 23 24

```

```

1 0 0 0 0 0 0 1 1 0 1 1
2 0 1 1 1 1 0 0 0 0 0 1
3 6 7 8 0 0 1 1 1 0 1 1
4 5 1 9 1 0 0 17 18 19 1 1
0 0 1 10 11 1 1 16 1 20 1 1
0 0 0 12 13 14 15 1 21 22 1
0 1 1 1 1 1 1 1 1 1 23 24

```

```

1 0 0 0 0 0 0 1 1 0 1 1
2 0 1 1 1 1 0 0 0 0 0 1
3 6 7 8 9 10 1 1 1 0 1 1
4 5 1 0 1 11 12 13 14 15 1 1
0 0 1 0 0 1 1 0 1 16 1 1
0 0 0 0 0 0 0 0 1 17 18 1
0 1 1 1 1 1 1 1 1 1 19 20

```

```

1 8 9 10 11 12 13 1 1 0 1 1
2 7 1 1 1 1 14 15 16 17 0 1
3 6 0 0 0 0 1 1 1 18 1 1
4 5 1 0 1 0 0 0 0 19 1 1
0 0 1 0 0 1 1 0 1 20 1 1
0 0 0 0 0 0 0 0 1 21 22 1
0 1 1 1 1 1 1 1 1 1 23 24

```

```

1 0 0 0 0 0 0 1 1 0 1 1
2 0 1 1 1 1 0 0 0 0 0 1
3 4 0 0 0 0 1 1 1 0 1 1
0 5 1 0 1 0 0 15 16 17 1 1
0 6 1 0 0 1 1 14 1 18 1 1
0 7 8 9 10 11 12 13 1 19 20 1
0 1 1 1 1 1 1 1 1 1 21 22

```

```

1 0 0 0 0 0 0 1 1 0 1 1
2 0 1 1 1 1 0 0 0 0 0 1
3 4 0 14 15 16 1 1 1 0 1 1
0 5 1 13 1 17 18 19 20 21 1 1
0 6 1 12 11 1 1 0 1 22 1 1
0 7 8 9 10 0 0 0 1 23 24 1
0 1 1 1 1 1 1 1 1 1 25 26

```

```

1 0 0 0 0 0 0 1 1 0 1 1
2 0 1 1 1 1 0 0 0 0 0 1
3 4 0 0 0 0 1 1 1 0 1 1
0 5 1 0 1 0 0 17 18 19 1 1

```



```

0 6 1 10 11 1 1 16 1 20 1 1
0 7 8 9 12 13 14 15 1 21 22 1
0 1 1 1 1 1 1 1 1 1 1 23 24

```

```

1 0 0 0 0 0 0 1 1 0 1 1
2 0 1 1 1 1 0 0 0 0 0 1
3 4 0 12 13 14 1 1 1 0 1 1
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0 6 1 10 0 1 1 0 1 20 1 1
0 7 8 9 0 0 0 0 1 21 22 1
0 1 1 1 1 1 1 1 1 1 1 23 24

```

```

1 0 0 0 0 0 0 1 1 0 1 1
2 0 1 1 1 1 0 0 0 0 0 1
3 4 0 0 0 0 1 1 1 0 1 1
0 5 1 0 1 0 0 17 18 19 1 1
7 6 1 0 0 1 1 16 1 20 1 1
8 9 10 11 12 13 14 15 1 21 22 1
0 1 1 1 1 1 1 1 1 1 1 23 24

```

```

1 0 0 0 0 0 0 1 1 0 1 1
2 0 1 1 1 1 0 0 0 0 0 1
3 4 0 16 17 18 1 1 1 0 1 1
0 5 1 15 1 19 20 21 22 23 1 1
7 6 1 14 13 1 1 0 1 24 1 1
8 9 10 11 12 0 0 0 1 25 26 1
0 1 1 1 1 1 1 1 1 1 1 27 28

```

```

1 0 0 0 0 0 0 1 1 0 1 1
2 0 1 1 1 1 0 0 0 0 0 1
3 4 0 0 0 0 1 1 1 0 1 1
0 5 1 0 1 0 0 19 20 21 1 1
7 6 1 12 13 1 1 18 1 22 1 1
8 9 10 11 14 15 16 17 1 23 24 1
0 1 1 1 1 1 1 1 1 1 1 25 26

```

```

1 0 0 0 0 0 0 1 1 0 1 1
2 0 1 1 1 1 0 0 0 0 0 1
3 4 0 14 15 16 1 1 1 0 1 1
0 5 1 13 1 17 18 19 20 21 1 1
7 6 1 12 0 1 1 0 1 22 1 1
8 9 10 11 0 0 0 0 1 23 24 1
0 1 1 1 1 1 1 1 1 1 1 25 26

```

```

1 0 0 0 0 0 0 1 1 0 1 1
2 0 1 1 1 1 0 0 0 0 0 1
3 4 0 0 0 0 1 1 1 0 1 1
6 5 1 0 1 0 0 17 18 19 1 1
7 0 1 0 0 1 1 16 1 20 1 1
8 9 10 11 12 13 14 15 1 21 22 1
0 1 1 1 1 1 1 1 1 1 1 23 24

```

```

1 0 0 0 0 0 0 1 1 0 1 1
2 0 1 1 1 1 0 0 0 0 0 1
3 4 0 16 17 18 1 1 1 0 1 1
6 5 1 15 1 19 20 21 22 23 1 1
7 0 1 14 13 1 1 0 1 24 1 1
8 9 10 11 12 0 0 0 1 25 26 1
0 1 1 1 1 1 1 1 1 1 1 27 28

```

```

1 0 0 0 0 0 0 1 1 0 1 1
2 0 1 1 1 1 0 0 0 0 0 1
3 4 0 0 0 0 1 1 1 0 1 1
6 5 1 0 1 0 0 19 20 21 1 1
7 0 1 12 13 1 1 18 1 22 1 1
8 9 10 11 14 15 16 17 1 23 24 1
0 1 1 1 1 1 1 1 1 1 1 25 26

```

```

1 0 0 0 0 0 0 1 1 0 1 1

```

```

2 0 1 1 1 1 0 0 0 0 0 1
3 4 0 14 15 16 1 1 1 0 1 1
6 5 1 13 1 17 18 19 20 21 1 1
7 0 1 12 0 1 1 0 1 22 1 1
8 9 10 11 0 0 0 0 1 23 24 1
0 1 1 1 1 1 1 1 1 1 25 26

```

```

1 0 0 0 0 0 0 1 1 0 1 1
2 0 1 1 1 1 0 0 0 0 0 1
3 4 0 0 0 0 1 1 1 0 1 1
6 5 1 0 1 0 0 17 18 19 1 1
7 8 1 0 0 1 1 16 1 20 1 1
0 9 10 11 12 13 14 15 1 21 22 1
0 1 1 1 1 1 1 1 1 1 23 24

```

```

1 0 0 0 0 0 0 1 1 0 1 1
2 0 1 1 1 1 0 0 0 0 0 1
3 4 0 16 17 18 1 1 1 0 1 1
6 5 1 15 1 19 20 21 22 23 1 1
7 8 1 14 13 1 1 0 1 24 1 1
0 9 10 11 12 0 0 0 1 25 26 1
0 1 1 1 1 1 1 1 1 1 27 28

```

```

1 0 0 0 0 0 0 1 1 0 1 1
2 0 1 1 1 1 0 0 0 0 0 1
3 4 0 0 0 0 1 1 1 0 1 1
6 5 1 0 1 0 0 19 20 21 1 1
7 8 1 12 13 1 1 18 1 22 1 1
0 9 10 11 14 15 16 17 1 23 24 1
0 1 1 1 1 1 1 1 1 1 25 26

```

```

1 0 0 0 0 0 0 1 1 0 1 1
2 0 1 1 1 1 0 0 0 0 0 1
3 4 0 14 15 16 1 1 1 0 1 1
6 5 1 13 1 17 18 19 20 21 1 1
7 8 1 12 0 1 1 0 1 22 1 1
0 9 10 11 0 0 0 0 1 23 24 1
0 1 1 1 1 1 1 1 1 1 25 26

```

```

1 0 0 0 0 0 0 1 1 0 1 1
2 0 1 1 1 1 0 0 0 0 0 1
3 4 5 6 0 0 1 1 1 0 1 1
0 0 1 7 1 0 0 15 16 17 1 1
0 0 1 8 0 1 1 14 1 18 1 1
0 0 0 9 10 11 12 13 1 19 20 1
0 1 1 1 1 1 1 1 1 1 21 22

```

```

1 0 0 0 0 0 0 1 1 0 1 1
2 0 1 1 1 1 0 0 0 0 0 1
3 4 5 6 0 0 1 1 1 0 1 1
0 0 1 7 1 0 0 15 16 17 1 1
0 0 1 8 9 1 1 14 1 18 1 1
0 0 0 0 10 11 12 13 1 19 20 1
0 1 1 1 1 1 1 1 1 1 21 22

```

```

1 0 0 0 0 0 0 1 1 0 1 1
2 0 1 1 1 1 0 0 0 0 0 1
3 4 5 6 7 8 1 1 1 0 1 1
0 0 1 0 1 9 10 11 12 13 1 1
0 0 1 0 0 1 1 0 1 14 1 1
0 0 0 0 0 0 0 0 1 15 16 1
0 1 1 1 1 1 1 1 1 1 17 18

```

```

1 6 7 8 9 10 11 1 1 0 1 1
2 5 1 1 1 1 12 13 14 15 0 1
3 4 0 0 0 0 1 1 1 16 1 1
0 0 1 0 1 0 0 0 0 17 1 1
0 0 1 0 0 1 1 0 1 18 1 1
0 0 0 0 0 0 0 0 1 19 20 1

```

```
0  1  1  1  1  1  1  1  1  1  1 21 22
```

```
1  0  0  0  0  0  0  1  1  0  1  1
2  3  1  1  1  1  0  0  0  0  0  1
0  4  0  0  0  0  1  1  1  0  1  1
0  5  1  0  1  0  0 15 16 17  1  1
0  6  1  0  0  1  1 14  1 18  1  1
0  7  8  9 10 11 12 13  1 19 20  1
0  1  1  1  1  1  1  1  1  1  1 21 22
```

```
1  0  0  0  0  0  0  1  1  0  1  1
2  3  1  1  1  1  0  0  0  0  0  1
0  4  0 14 15 16  1  1  1  0  1  1
0  5  1 13  1 17 18 19 20 21  1  1
0  6  1 12 11  1  1  0  1 22  1  1
0  7  8  9 10  0  0  0  1 23 24  1
0  1  1  1  1  1  1  1  1  1  1 25 26
```

```
1  0  0  0  0  0  0  1  1  0  1  1
2  3  1  1  1  1  0  0  0  0  0  1
0  4  0  0  0  0  1  1  1  0  1  1
0  5  1  0  1  0  0 17 18 19  1  1
0  6  1 10 11  1  1 16  1 20  1  1
0  7  8  9 12 13 14 15  1 21 22  1
0  1  1  1  1  1  1  1  1  1  1 23 24
```

```
1  0  0  0  0  0  0  1  1  0  1  1
2  3  1  1  1  1  0  0  0  0  0  1
0  4  0 12 13 14  1  1  1  0  1  1
0  5  1 11  1 15 16 17 18 19  1  1
0  6  1 10  0  1  1  0  1 20  1  1
0  7  8  9  0  0  0  0  1 21 22  1
0  1  1  1  1  1  1  1  1  1  1 23 24
```

```
1  0  0  0  0  0  0  1  1  0  1  1
2  3  1  1  1  1  0  0  0  0  0  1
0  4  0  0  0  0  1  1  1  0  1  1
0  5  1  0  1  0  0 17 18 19  1  1
7  6  1  0  0  1  1 16  1 20  1  1
8  9 10 11 12 13 14 15  1 21 22  1
0  1  1  1  1  1  1  1  1  1  1 23 24
```

```
1  0  0  0  0  0  0  1  1  0  1  1
2  3  1  1  1  1  0  0  0  0  0  1
0  4  0 16 17 18  1  1  1  0  1  1
0  5  1 15  1 19 20 21 22 23  1  1
7  6  1 14 13  1  1  0  1 24  1  1
8  9 10 11 12  0  0  0  1 25 26  1
0  1  1  1  1  1  1  1  1  1  1 27 28
```

```
1  0  0  0  0  0  0  1  1  0  1  1
2  3  1  1  1  1  0  0  0  0  0  1
0  4  0  0  0  0  1  1  1  0  1  1
0  5  1  0  1  0  0 19 20 21  1  1
7  6  1 12 13  1  1 18  1 22  1  1
8  9 10 11 14 15 16 17  1 23 24  1
0  1  1  1  1  1  1  1  1  1  1 25 26
```

```
1  0  0  0  0  0  0  1  1  0  1  1
2  3  1  1  1  1  0  0  0  0  0  1
0  4  0 14 15 16  1  1  1  0  1  1
0  5  1 13  1 17 18 19 20 21  1  1
7  6  1 12  0  1  1  0  1 22  1  1
8  9 10 11  0  0  0  0  1 23 24  1
0  1  1  1  1  1  1  1  1  1  1 25 26
```

```
1  0  0  0  0  0  0  1  1  0  1  1
2  3  1  1  1  1  0  0  0  0  0  1
0  4  0  0  0  0  1  1  1  0  1  1
```

```

6 5 1 0 1 0 0 17 18 19 1 1
7 0 1 0 0 1 1 16 1 20 1 1
8 9 10 11 12 13 14 15 1 21 22 1
0 1 1 1 1 1 1 1 1 1 23 24

```

```

1 0 0 0 0 0 0 1 1 0 1 1
2 3 1 1 1 1 0 0 0 0 0 1
0 4 0 16 17 18 1 1 1 0 1 1
6 5 1 15 1 19 20 21 22 23 1 1
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```

```

1 0 0 0 0 0 0 1 1 0 1 1
2 3 1 1 1 1 0 0 0 0 0 1
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```

```

1 0 0 0 0 0 0 1 1 0 1 1
2 3 1 1 1 1 0 0 0 0 0 1
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```

```

1 0 0 0 0 0 0 1 1 0 1 1
2 3 1 1 1 1 0 0 0 0 0 1
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```

```

1 0 0 0 0 0 0 1 1 0 1 1
2 3 1 1 1 1 0 0 0 0 0 1
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```

```

1 0 0 0 0 0 0 1 1 0 1 1
2 3 1 1 1 1 0 0 0 0 0 1
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```

```

1 0 0 0 0 0 0 1 1 0 1 1
2 3 1 1 1 1 0 0 0 0 0 1
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```

```

1 0 0 0 0 0 0 1 1 0 1 1
2 3 1 1 1 1 0 0 0 0 0 1
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0 0 0 9 10 11 12 13 1 19 20 1
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```

```

1 0 0 0 0 0 0 1 1 0 1 1
2 3 1 1 1 1 0 0 0 0 0 1
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0 0 1 7 1 0 0 15 16 17 1 1
0 0 1 8 9 1 1 14 1 18 1 1
0 0 0 0 10 11 12 13 1 19 20 1
0 1 1 1 1 1 1 1 1 1 1 21 22

```

```

1 0 0 0 0 0 0 1 1 0 1 1
2 3 1 1 1 1 0 0 0 0 0 1
0 4 5 6 7 8 1 1 1 0 1 1
0 0 1 0 1 9 10 11 12 13 1 1
0 0 1 0 0 1 1 0 1 14 1 1
0 0 0 0 0 0 0 0 1 15 16 1
0 1 1 1 1 1 1 1 1 1 1 17 18

```

```

1 0 0 0 0 0 0 1 1 0 1 1
2 3 1 1 1 1 0 0 0 0 0 1
5 4 0 0 0 0 1 1 1 0 1 1
6 0 1 0 1 0 0 17 18 19 1 1
7 0 1 0 0 1 1 16 1 20 1 1
8 9 10 11 12 13 14 15 1 21 22 1
0 1 1 1 1 1 1 1 1 1 1 23 24

```

```

1 0 0 0 0 0 0 1 1 0 1 1
2 3 1 1 1 1 0 0 0 0 0 1
5 4 0 16 17 18 1 1 1 0 1 1
6 0 1 15 1 19 20 21 22 23 1 1
7 0 1 14 13 1 1 0 1 24 1 1
8 9 10 11 12 0 0 0 1 25 26 1
0 1 1 1 1 1 1 1 1 1 1 27 28

```

```

1 0 0 0 0 0 0 1 1 0 1 1
2 3 1 1 1 1 0 0 0 0 0 1
5 4 0 0 0 0 1 1 1 0 1 1
6 0 1 0 1 0 0 19 20 21 1 1
7 0 1 12 13 1 1 18 1 22 1 1
8 9 10 11 14 15 16 17 1 23 24 1
0 1 1 1 1 1 1 1 1 1 1 25 26

```

```

1 0 0 0 0 0 0 1 1 0 1 1
2 3 1 1 1 1 0 0 0 0 0 1
5 4 0 14 15 16 1 1 1 0 1 1
6 0 1 13 1 17 18 19 20 21 1 1
7 0 1 12 0 1 1 0 1 22 1 1
8 9 10 11 0 0 0 0 1 23 24 1
0 1 1 1 1 1 1 1 1 1 1 25 26

```

```

1 0 0 0 0 0 0 1 1 0 1 1
2 3 1 1 1 1 0 0 0 0 0 1
5 4 0 0 0 0 1 1 1 0 1 1
6 0 1 0 1 0 0 17 18 19 1 1
7 8 1 0 0 1 1 16 1 20 1 1
0 9 10 11 12 13 14 15 1 21 22 1
0 1 1 1 1 1 1 1 1 1 1 23 24

```

```

1 0 0 0 0 0 0 1 1 0 1 1
2 3 1 1 1 1 0 0 0 0 0 1
5 4 0 16 17 18 1 1 1 0 1 1
6 0 1 15 1 19 20 21 22 23 1 1
7 8 1 14 13 1 1 0 1 24 1 1
0 9 10 11 12 0 0 0 1 25 26 1
0 1 1 1 1 1 1 1 1 1 1 27 28

```

```

1 0 0 0 0 0 0 1 1 0 1 1
2 3 1 1 1 1 0 0 0 0 0 1
5 4 0 0 0 0 1 1 1 0 1 1
6 0 1 0 1 0 0 19 20 21 1 1
7 8 1 12 13 1 1 18 1 22 1 1

```

```

0  9 10 11 14 15 16 17  1 23 24  1
0  1  1  1  1  1  1  1  1  1  1 25 26

1  0  0  0  0  0  0  1  1  0  1  1
2  3  1  1  1  1  0  0  0  0  0  1
5  4  0 14 15 16  1  1  1  0  1  1
6  0  1 13  1 17 18 19 20 21  1  1
7  8  1 12  0  1  1  0  1 22  1  1
0  9 10 11  0  0  0  0  1 23 24  1
0  1  1  1  1  1  1  1  1  1  1 25 26

1  0  0  0  0  0  0  1  1  0  1  1
2  3  1  1  1  1  0  0  0  0  0  1
5  4  0  0  0  0  1  1  1  0  1  1
6  7  1  0  1  0  0 17 18 19  1  1
0  8  1  0  0  1  1 16  1 20  1  1
0  9 10 11 12 13 14 15  1 21 22  1
0  1  1  1  1  1  1  1  1  1  1 23 24

1  0  0  0  0  0  0  1  1  0  1  1
2  3  1  1  1  1  0  0  0  0  0  1
5  4  0 16 17 18  1  1  1  0  1  1
6  7  1 15  1 19 20 21 22 23  1  1
0  8  1 14 13  1  1  0  1 24  1  1
0  9 10 11 12  0  0  0  1 25 26  1
0  1  1  1  1  1  1  1  1  1  1 27 28

1  0  0  0  0  0  0  1  1  0  1  1
2  3  1  1  1  1  0  0  0  0  0  1
5  4  0  0  0  0  1  1  1  0  1  1
6  7  1  0  1  0  0 19 20 21  1  1
0  8  1 12 13  1  1 18  1 22  1  1
0  9 10 11 14 15 16 17  1 23 24  1
0  1  1  1  1  1  1  1  1  1  1 25 26

1  0  0  0  0  0  0  1  1  0  1  1
2  3  1  1  1  1  0  0  0  0  0  1
5  4  0 14 15 16  1  1  1  0  1  1
6  7  1 13  1 17 18 19 20 21  1  1
0  8  1 12  0  1  1  0  1 22  1  1
0  9 10 11  0  0  0  0  1 23 24  1
0  1  1  1  1  1  1  1  1  1  1 25 26

1  0  0  0  0  0  0  1  1  0  1  1
2  3  1  1  1  1  0  0  0  0  0  1
5  4  0  0  0  0  1  1  1  0  1  1
6  7  1  0  1  0  0 19 20 21  1  1
9  8  1  0  0  1  1 18  1 22  1  1
10 11 12 13 14 15 16 17  1 23 24  1
0  1  1  1  1  1  1  1  1  1  1 25 26

1  0  0  0  0  0  0  1  1  0  1  1
2  3  1  1  1  1  0  0  0  0  0  1
5  4  0 18 19 20  1  1  1  0  1  1
6  7  1 17  1 21 22 23 24 25  1  1
9  8  1 16 15  1  1  0  1 26  1  1
10 11 12 13 14  0  0  0  1 27 28  1
0  1  1  1  1  1  1  1  1  1  1 29 30

1  0  0  0  0  0  0  1  1  0  1  1
2  3  1  1  1  1  0  0  0  0  0  1
5  4  0  0  0  0  1  1  1  0  1  1
6  7  1  0  1  0  0 21 22 23  1  1
9  8  1 14 15  1  1 20  1 24  1  1
10 11 12 13 16 17 18 19  1 25 26  1
0  1  1  1  1  1  1  1  1  1  1 27 28

1  0  0  0  0  0  0  1  1  0  1  1
2  3  1  1  1  1  0  0  0  0  0  1

```

```

5  4  0 16 17 18  1  1  1  0  1  1
6  7  1 15  1 19 20 21 22 23  1  1
9  8  1 14  0  1  1  0  1 24  1  1
10 11 12 13  0  0  0  0  1 25 26  1
0  1  1  1  1  1  1  1  1  1 27 28

```

```

1  4  5  6  7  8  9  1  1  0  1  1
2  3  1  1  1  1 10 11 12 13  0  1
0  0  0  0  0  0  1  1  1 14  1  1
0  0  1  0  1  0  0  0  0 15  1  1
0  0  1  0  0  1  1  0  1 16  1  1
0  0  0  0  0  0  0  0  1 17 18  1
0  1  1  1  1  1  1  1  1  1 19 20

```

```

1  2  0  0  0  0  0  1  1  0  1  1
0  3  1  1  1  1  0  0  0  0  0  1
0  4  0  0  0  0  0  1  1  1  0  1  1
0  5  1  0  1  0  0 15 16 17  1  1
0  6  1  0  0  1  1 14  1 18  1  1
0  7  8  9 10 11 12 13  1 19 20  1
0  1  1  1  1  1  1  1  1  1 21 22

```

```

1  2  0  0  0  0  0  1  1  0  1  1
0  3  1  1  1  1  0  0  0  0  0  1
0  4  0 14 15 16  1  1  1  0  1  1
0  5  1 13  1 17 18 19 20 21  1  1
0  6  1 12 11  1  1  0  1 22  1  1
0  7  8  9 10  0  0  0  1 23 24  1
0  1  1  1  1  1  1  1  1  1 25 26

```

```

1  2  0  0  0  0  0  1  1  0  1  1
0  3  1  1  1  1  0  0  0  0  0  1
0  4  0  0  0  0  0  1  1  1  0  1  1
0  5  1  0  1  0  0 17 18 19  1  1
0  6  1 10 11  1  1 16  1 20  1  1
0  7  8  9 12 13 14 15  1 21 22  1
0  1  1  1  1  1  1  1  1  1 23 24

```

```

1  2  0  0  0  0  0  1  1  0  1  1
0  3  1  1  1  1  0  0  0  0  0  1
0  4  0 12 13 14  1  1  1  0  1  1
0  5  1 11  1 15 16 17 18 19  1  1
0  6  1 10  0  1  1  0  1 20  1  1
0  7  8  9  0  0  0  0  1 21 22  1
0  1  1  1  1  1  1  1  1  1 23 24

```

```

1  2  0  0  0  0  0  1  1  0  1  1
0  3  1  1  1  1  0  0  0  0  0  1
0  4  0  0  0  0  0  1  1  1  0  1  1
0  5  1  0  1  0  0 17 18 19  1  1
7  6  1  0  0  1  1 16  1 20  1  1
8  9 10 11 12 13 14 15  1 21 22  1
0  1  1  1  1  1  1  1  1  1 23 24

```

```

1  2  0  0  0  0  0  1  1  0  1  1
0  3  1  1  1  1  0  0  0  0  0  1
0  4  0 16 17 18  1  1  1  0  1  1
0  5  1 15  1 19 20 21 22 23  1  1
7  6  1 14 13  1  1  0  1 24  1  1
8  9 10 11 12  0  0  0  1 25 26  1
0  1  1  1  1  1  1  1  1  1 27 28

```

```

1  2  0  0  0  0  0  1  1  0  1  1
0  3  1  1  1  1  0  0  0  0  0  1
0  4  0  0  0  0  0  1  1  1  0  1  1
0  5  1  0  1  0  0 19 20 21  1  1
7  6  1 12 13  1  1 18  1 22  1  1
8  9 10 11 14 15 16 17  1 23 24  1
0  1  1  1  1  1  1  1  1  1 25 26

```

```

1 2 0 0 0 0 0 1 1 0 1 1
0 3 1 1 1 1 0 0 0 0 0 1
0 4 0 14 15 16 1 1 1 0 1 1
0 5 1 13 1 17 18 19 20 21 1 1
7 6 1 12 0 1 1 0 1 22 1 1
8 9 10 11 0 0 0 0 1 23 24 1
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```

```

1 2 0 0 0 0 0 1 1 0 1 1
0 3 1 1 1 1 0 0 0 0 0 1
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6 5 1 0 1 0 0 17 18 19 1 1
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```

```

1 2 0 0 0 0 0 1 1 0 1 1
0 3 1 1 1 1 0 0 0 0 0 1
0 4 0 16 17 18 1 1 1 0 1 1
6 5 1 15 1 19 20 21 22 23 1 1
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```

```

1 2 0 0 0 0 0 1 1 0 1 1
0 3 1 1 1 1 0 0 0 0 0 1
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6 5 1 0 1 0 0 19 20 21 1 1
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```

```

1 2 0 0 0 0 0 1 1 0 1 1
0 3 1 1 1 1 0 0 0 0 0 1
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```

```

1 2 0 0 0 0 0 1 1 0 1 1
0 3 1 1 1 1 0 0 0 0 0 1
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```

```

1 2 0 0 0 0 0 1 1 0 1 1
0 3 1 1 1 1 0 0 0 0 0 1
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```

```

1 2 0 0 0 0 0 1 1 0 1 1
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```

```

1 2 0 0 0 0 0 1 1 0 1 1
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```



```

7 8 1 12 0 1 1 0 1 22 1 1
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```

```

1 2 0 0 0 0 0 1 1 0 1 1
0 3 1 1 1 1 0 0 0 0 0 1
0 4 5 6 0 0 1 1 1 0 1 1
0 0 1 7 1 0 0 15 16 17 1 1
0 0 1 8 0 1 1 14 1 18 1 1
0 0 0 9 10 11 12 13 1 19 20 1
0 1 1 1 1 1 1 1 1 1 21 22

```

```

1 2 0 0 0 0 0 1 1 0 1 1
0 3 1 1 1 1 0 0 0 0 0 1
0 4 5 6 0 0 1 1 1 0 1 1
0 0 1 7 1 0 0 15 16 17 1 1
0 0 1 8 9 1 1 14 1 18 1 1
0 0 0 0 10 11 12 13 1 19 20 1
0 1 1 1 1 1 1 1 1 1 21 22

```

```

1 2 0 0 0 0 0 1 1 0 1 1
0 3 1 1 1 1 0 0 0 0 0 1
0 4 5 6 7 8 1 1 1 0 1 1
0 0 1 0 1 9 10 11 12 13 1 1
0 0 1 0 0 1 1 0 1 14 1 1
0 0 0 0 0 0 0 0 1 15 16 1
0 1 1 1 1 1 1 1 1 1 17 18

```

```

1 2 0 0 0 0 0 1 1 0 1 1
0 3 1 1 1 1 0 0 0 0 0 1
5 4 0 0 0 0 1 1 1 0 1 1
6 0 1 0 1 0 0 17 18 19 1 1
7 0 1 0 0 1 1 16 1 20 1 1
8 9 10 11 12 13 14 15 1 21 22 1
0 1 1 1 1 1 1 1 1 1 23 24

```

```

1 2 0 0 0 0 0 1 1 0 1 1
0 3 1 1 1 1 0 0 0 0 0 1
5 4 0 16 17 18 1 1 1 0 1 1
6 0 1 15 1 19 20 21 22 23 1 1
7 0 1 14 13 1 1 0 1 24 1 1
8 9 10 11 12 0 0 0 1 25 26 1
0 1 1 1 1 1 1 1 1 1 27 28

```

```

1 2 0 0 0 0 0 1 1 0 1 1
0 3 1 1 1 1 0 0 0 0 0 1
5 4 0 0 0 0 1 1 1 0 1 1
6 0 1 0 1 0 0 19 20 21 1 1
7 0 1 12 13 1 1 18 1 22 1 1
8 9 10 11 14 15 16 17 1 23 24 1
0 1 1 1 1 1 1 1 1 1 25 26

```

```

1 2 0 0 0 0 0 1 1 0 1 1
0 3 1 1 1 1 0 0 0 0 0 1
5 4 0 14 15 16 1 1 1 0 1 1
6 0 1 13 1 17 18 19 20 21 1 1
7 0 1 12 0 1 1 0 1 22 1 1
8 9 10 11 0 0 0 0 1 23 24 1
0 1 1 1 1 1 1 1 1 1 25 26

```

```

1 2 0 0 0 0 0 1 1 0 1 1
0 3 1 1 1 1 0 0 0 0 0 1
5 4 0 0 0 0 1 1 1 0 1 1
6 0 1 0 1 0 0 17 18 19 1 1
7 8 1 0 0 1 1 16 1 20 1 1
0 9 10 11 12 13 14 15 1 21 22 1
0 1 1 1 1 1 1 1 1 1 23 24

```

```

1 2 0 0 0 0 0 1 1 0 1 1

```

```

0 3 1 1 1 1 0 0 0 0 0 1
5 4 0 16 17 18 1 1 1 0 1 1
6 0 1 15 1 19 20 21 22 23 1 1
7 8 1 14 13 1 1 0 1 24 1 1
0 9 10 11 12 0 0 0 1 25 26 1
0 1 1 1 1 1 1 1 1 1 27 28

```

```

1 2 0 0 0 0 0 1 1 0 1 1
0 3 1 1 1 1 0 0 0 0 0 1
5 4 0 0 0 0 1 1 1 0 1 1
6 0 1 0 1 0 0 19 20 21 1 1
7 8 1 12 13 1 1 18 1 22 1 1
0 9 10 11 14 15 16 17 1 23 24 1
0 1 1 1 1 1 1 1 1 1 25 26

```

```

1 2 0 0 0 0 0 1 1 0 1 1
0 3 1 1 1 1 0 0 0 0 0 1
5 4 0 14 15 16 1 1 1 0 1 1
6 0 1 13 1 17 18 19 20 21 1 1
7 8 1 12 0 1 1 0 1 22 1 1
0 9 10 11 0 0 0 0 1 23 24 1
0 1 1 1 1 1 1 1 1 1 25 26

```

```

1 2 0 0 0 0 0 1 1 0 1 1
0 3 1 1 1 1 0 0 0 0 0 1
5 4 0 0 0 0 1 1 1 0 1 1
6 7 1 0 1 0 0 17 18 19 1 1
0 8 1 0 0 1 1 16 1 20 1 1
0 9 10 11 12 13 14 15 1 21 22 1
0 1 1 1 1 1 1 1 1 1 23 24

```

```

1 2 0 0 0 0 0 1 1 0 1 1
0 3 1 1 1 1 0 0 0 0 0 1
5 4 0 16 17 18 1 1 1 0 1 1
6 7 1 15 1 19 20 21 22 23 1 1
0 8 1 14 13 1 1 0 1 24 1 1
0 9 10 11 12 0 0 0 1 25 26 1
0 1 1 1 1 1 1 1 1 1 27 28

```

```

1 2 0 0 0 0 0 1 1 0 1 1
0 3 1 1 1 1 0 0 0 0 0 1
5 4 0 0 0 0 1 1 1 0 1 1
6 7 1 0 1 0 0 19 20 21 1 1
0 8 1 12 13 1 1 18 1 22 1 1
0 9 10 11 14 15 16 17 1 23 24 1
0 1 1 1 1 1 1 1 1 1 25 26

```

```

1 2 0 0 0 0 0 1 1 0 1 1
0 3 1 1 1 1 0 0 0 0 0 1
5 4 0 14 15 16 1 1 1 0 1 1
6 7 1 13 1 17 18 19 20 21 1 1
0 8 1 12 0 1 1 0 1 22 1 1
0 9 10 11 0 0 0 0 1 23 24 1
0 1 1 1 1 1 1 1 1 1 25 26

```

```

1 2 0 0 0 0 0 1 1 0 1 1
0 3 1 1 1 1 0 0 0 0 0 1
5 4 0 0 0 0 1 1 1 0 1 1
6 7 1 0 1 0 0 19 20 21 1 1
9 8 1 0 0 1 1 18 1 22 1 1
10 11 12 13 14 15 16 17 1 23 24 1
0 1 1 1 1 1 1 1 1 1 25 26

```

```

1 2 0 0 0 0 0 1 1 0 1 1
0 3 1 1 1 1 0 0 0 0 0 1
5 4 0 18 19 20 1 1 1 0 1 1
6 7 1 17 1 21 22 23 24 25 1 1
9 8 1 16 15 1 1 0 1 26 1 1
10 11 12 13 14 0 0 0 1 27 28 1

```

```

0  1  1  1  1  1  1  1  1  1  1 29 30

1  2  0  0  0  0  0  1  1  0  1  1
0  3  1  1  1  1  0  0  0  0  0  1
5  4  0  0  0  0  1  1  1  0  1  1
6  7  1  0  1  0  0 21 22 23  1  1
9  8  1 14 15  1  1 20  1 24  1  1
10 11 12 13 16 17 18 19  1 25 26  1
0  1  1  1  1  1  1  1  1  1  1 27 28

1  2  0  0  0  0  0  1  1  0  1  1
0  3  1  1  1  1  0  0  0  0  0  1
5  4  0 16 17 18  1  1  1  0  1  1
6  7  1 15  1 19 20 21 22 23  1  1
9  8  1 14  0  1  1  0  1 24  1  1
10 11 12 13  0  0  0  0  1 25 26  1
0  1  1  1  1  1  1  1  1  1  1 27 28

1  2  0  0  0  0  0  1  1  0  1  1
4  3  1  1  1  1  0  0  0  0  0  1
5  0  0  0  0  0  1  1  1  0  1  1
6  0  1  0  1  0  0 17 18 19  1  1
7  0  1  0  0  1  1 16  1 20  1  1
8  9 10 11 12 13 14 15  1 21 22  1
0  1  1  1  1  1  1  1  1  1  1 23 24

1  2  0  0  0  0  0  1  1  0  1  1
4  3  1  1  1  1  0  0  0  0  0  1
5  0  0 16 17 18  1  1  1  0  1  1
6  0  1 15  1 19 20 21 22 23  1  1
7  0  1 14 13  1  1  0  1 24  1  1
8  9 10 11 12  0  0  0  1 25 26  1
0  1  1  1  1  1  1  1  1  1  1 27 28

1  2  0  0  0  0  0  1  1  0  1  1
4  3  1  1  1  1  0  0  0  0  0  1
5  0  0  0  0  0  1  1  1  0  1  1
6  0  1  0  1  0  0 19 20 21  1  1
7  0  1 12 13  1  1 18  1 22  1  1
8  9 10 11 14 15 16 17  1 23 24  1
0  1  1  1  1  1  1  1  1  1  1 25 26

1  2  0  0  0  0  0  1  1  0  1  1
4  3  1  1  1  1  0  0  0  0  0  1
5  0  0 14 15 16  1  1  1  0  1  1
6  0  1 13  1 17 18 19 20 21  1  1
7  0  1 12  0  1  1  0  1 22  1  1
8  9 10 11  0  0  0  0  1 23 24  1
0  1  1  1  1  1  1  1  1  1  1 25 26

1  2  0  0  0  0  0  1  1  0  1  1
4  3  1  1  1  1  0  0  0  0  0  1
5 12 13 14  0  0  1  1  1  0  1  1
6 11  1 15  1  0  0 23 24 25  1  1
7 10  1 16  0  1  1 22  1 26  1  1
8  9  0 17 18 19 20 21  1 27 28  1
0  1  1  1  1  1  1  1  1  1  1 29 30

1  2  0  0  0  0  0  1  1  0  1  1
4  3  1  1  1  1  0  0  0  0  0  1
5 12 13 14  0  0  1  1  1  0  1  1
6 11  1 15  1  0  0 23 24 25  1  1
7 10  1 16 17  1  1 22  1 26  1  1
8  9  0  0 18 19 20 21  1 27 28  1
0  1  1  1  1  1  1  1  1  1  1 29 30

1  2  0  0  0  0  0  1  1  0  1  1
4  3  1  1  1  1  0  0  0  0  0  1
5 12 13 14 15 16  1  1  1  0  1  1

```

```

6 11 1 0 1 17 18 19 20 21 1 1
7 10 1 0 0 1 1 0 1 22 1 1
8 9 0 0 0 0 0 0 1 23 24 1
0 1 1 1 1 1 1 1 1 1 25 26

```

```

1 2 0 0 0 0 0 1 1 0 1 1
4 3 1 1 1 1 0 0 0 0 0 1
5 0 0 0 0 0 1 1 1 0 1 1
6 0 1 0 1 0 0 17 18 19 1 1
7 8 1 0 0 1 1 16 1 20 1 1
0 9 10 11 12 13 14 15 1 21 22 1
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```

```

1 2 0 0 0 0 0 1 1 0 1 1
4 3 1 1 1 1 0 0 0 0 0 1
5 0 0 16 17 18 1 1 1 0 1 1
6 0 1 15 1 19 20 21 22 23 1 1
7 8 1 14 13 1 1 0 1 24 1 1
0 9 10 11 12 0 0 0 1 25 26 1
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```

```

1 2 0 0 0 0 0 1 1 0 1 1
4 3 1 1 1 1 0 0 0 0 0 1
5 0 0 0 0 0 1 1 1 0 1 1
6 0 1 0 1 0 0 19 20 21 1 1
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```

```

1 2 0 0 0 0 0 1 1 0 1 1
4 3 1 1 1 1 0 0 0 0 0 1
5 0 0 14 15 16 1 1 1 0 1 1
6 0 1 13 1 17 18 19 20 21 1 1
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```

```

1 2 0 0 0 0 0 1 1 0 1 1
4 3 1 1 1 1 0 0 0 0 0 1
5 10 11 12 0 0 1 1 1 0 1 1
6 9 1 13 1 0 0 21 22 23 1 1
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```

```

1 2 0 0 0 0 0 1 1 0 1 1
4 3 1 1 1 1 0 0 0 0 0 1
5 10 11 12 0 0 1 1 1 0 1 1
6 9 1 13 1 0 0 21 22 23 1 1
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```

```

1 2 0 0 0 0 0 1 1 0 1 1
4 3 1 1 1 1 0 0 0 0 0 1
5 10 11 12 13 14 1 1 1 0 1 1
6 9 1 0 1 15 16 17 18 19 1 1
7 8 1 0 0 1 1 0 1 20 1 1
0 0 0 0 0 0 0 0 1 21 22 1
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```

```

1 2 0 0 0 0 0 1 1 0 1 1
4 3 1 1 1 1 0 0 0 0 0 1
5 0 0 0 0 0 1 1 1 0 1 1
6 7 1 0 1 0 0 17 18 19 1 1
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0 9 10 11 12 13 14 15 1 21 22 1
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```

```

1 2 0 0 0 0 0 1 1 0 1 1
4 3 1 1 1 1 0 0 0 0 0 1
5 0 0 16 17 18 1 1 1 0 1 1
6 7 1 15 1 19 20 21 22 23 1 1
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```

```

1 2 0 0 0 0 0 1 1 0 1 1
4 3 1 1 1 1 0 0 0 0 0 1
5 0 0 0 0 0 1 1 1 0 1 1
6 7 1 0 1 0 0 19 20 21 1 1
0 8 1 12 13 1 1 18 1 22 1 1
0 9 10 11 14 15 16 17 1 23 24 1
0 1 1 1 1 1 1 1 1 1 25 26

```

```

1 2 0 0 0 0 0 1 1 0 1 1
4 3 1 1 1 1 0 0 0 0 0 1
5 0 0 14 15 16 1 1 1 0 1 1
6 7 1 13 1 17 18 19 20 21 1 1
0 8 1 12 0 1 1 0 1 22 1 1
0 9 10 11 0 0 0 0 1 23 24 1
0 1 1 1 1 1 1 1 1 1 25 26

```

```

1 2 0 0 0 0 0 1 1 0 1 1
4 3 1 1 1 1 0 0 0 0 0 1
5 0 0 0 0 0 1 1 1 0 1 1
6 7 1 0 1 0 0 19 20 21 1 1
9 8 1 0 0 1 1 18 1 22 1 1
10 11 12 13 14 15 16 17 1 23 24 1
0 1 1 1 1 1 1 1 1 1 25 26

```

```

1 2 0 0 0 0 0 1 1 0 1 1
4 3 1 1 1 1 0 0 0 0 0 1
5 0 0 18 19 20 1 1 1 0 1 1
6 7 1 17 1 21 22 23 24 25 1 1
9 8 1 16 15 1 1 0 1 26 1 1
10 11 12 13 14 0 0 0 1 27 28 1
0 1 1 1 1 1 1 1 1 1 29 30

```

```

1 2 0 0 0 0 0 1 1 0 1 1
4 3 1 1 1 1 0 0 0 0 0 1
5 0 0 0 0 0 1 1 1 0 1 1
6 7 1 0 1 0 0 21 22 23 1 1
9 8 1 14 15 1 1 20 1 24 1 1
10 11 12 13 16 17 18 19 1 25 26 1
0 1 1 1 1 1 1 1 1 1 27 28

```

```

1 2 0 0 0 0 0 1 1 0 1 1
4 3 1 1 1 1 0 0 0 0 0 1
5 0 0 16 17 18 1 1 1 0 1 1
6 7 1 15 1 19 20 21 22 23 1 1
9 8 1 14 0 1 1 0 1 24 1 1
10 11 12 13 0 0 0 0 1 25 26 1
0 1 1 1 1 1 1 1 1 1 27 28

```

```

1 2 0 0 0 0 0 1 1 0 1 1
4 3 1 1 1 1 0 0 0 0 0 1
5 8 9 10 0 0 1 1 1 0 1 1
6 7 1 11 1 0 0 19 20 21 1 1
0 0 1 12 0 1 1 18 1 22 1 1
0 0 0 13 14 15 16 17 1 23 24 1
0 1 1 1 1 1 1 1 1 1 25 26

```

```

1 2 0 0 0 0 0 1 1 0 1 1
4 3 1 1 1 1 0 0 0 0 0 1
5 8 9 10 0 0 1 1 1 0 1 1
6 7 1 11 1 0 0 19 20 21 1 1
0 0 1 12 13 1 1 18 1 22 1 1

```

```

0 0 0 0 14 15 16 17 1 23 24 1
0 1 1 1 1 1 1 1 1 1 25 26

1 2 0 0 0 0 0 1 1 0 1 1
4 3 1 1 1 1 0 0 0 0 0 1
5 8 9 10 11 12 1 1 1 0 1 1
6 7 1 0 1 13 14 15 16 17 1 1
0 0 1 0 0 1 1 0 1 18 1 1
0 0 0 0 0 0 0 0 1 19 20 1
0 1 1 1 1 1 1 1 1 1 21 22

1 2 0 0 0 0 0 1 1 0 1 1
4 3 1 1 1 1 0 0 0 0 0 1
5 6 0 0 0 0 1 1 1 0 1 1
0 7 1 0 1 0 0 17 18 19 1 1
0 8 1 0 0 1 1 16 1 20 1 1
0 9 10 11 12 13 14 15 1 21 22 1
0 1 1 1 1 1 1 1 1 1 23 24

1 2 0 0 0 0 0 1 1 0 1 1
4 3 1 1 1 1 0 0 0 0 0 1
5 6 0 16 17 18 1 1 1 0 1 1
0 7 1 15 1 19 20 21 22 23 1 1
0 8 1 14 13 1 1 0 1 24 1 1
0 9 10 11 12 0 0 0 1 25 26 1
0 1 1 1 1 1 1 1 1 1 27 28

1 2 0 0 0 0 0 1 1 0 1 1
4 3 1 1 1 1 0 0 0 0 0 1
5 6 0 0 0 0 1 1 1 0 1 1
0 7 1 0 1 0 0 19 20 21 1 1
0 8 1 12 13 1 1 18 1 22 1 1
0 9 10 11 14 15 16 17 1 23 24 1
0 1 1 1 1 1 1 1 1 1 25 26

1 2 0 0 0 0 0 1 1 0 1 1
4 3 1 1 1 1 0 0 0 0 0 1
5 6 0 14 15 16 1 1 1 0 1 1
0 7 1 13 1 17 18 19 20 21 1 1
0 8 1 12 0 1 1 0 1 22 1 1
0 9 10 11 0 0 0 0 1 23 24 1
0 1 1 1 1 1 1 1 1 1 25 26

1 2 0 0 0 0 0 1 1 0 1 1
4 3 1 1 1 1 0 0 0 0 0 1
5 6 0 18 19 20 1 1 1 0 1 1
0 7 1 17 1 21 22 23 24 25 1 1
9 8 1 16 15 1 1 0 1 26 1 1
10 11 12 13 14 0 0 0 1 27 28 1
0 1 1 1 1 1 1 1 1 1 29 30

1 2 0 0 0 0 0 1 1 0 1 1
4 3 1 1 1 1 0 0 0 0 0 1
5 6 0 0 0 0 1 1 1 0 1 1
0 7 1 0 1 0 0 21 22 23 1 1
9 8 1 14 15 1 1 20 1 24 1 1
10 11 12 13 16 17 18 19 1 25 26 1
0 1 1 1 1 1 1 1 1 1 27 28

1 2 0 0 0 0 0 1 1 0 1 1
4 3 1 1 1 1 0 0 0 0 0 1

```

```

5 6 0 16 17 18 1 1 1 0 1 1
0 7 1 15 1 19 20 21 22 23 1 1
9 8 1 14 0 1 1 0 1 24 1 1
10 11 12 13 0 0 0 0 1 25 26 1
0 1 1 1 1 1 1 1 1 1 27 28

```

```

1 2 0 0 0 0 0 1 1 0 1 1
4 3 1 1 1 1 0 0 0 0 0 1
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8 7 1 0 1 0 0 19 20 21 1 1
9 0 1 0 0 1 1 18 1 22 1 1
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```

```

1 2 0 0 0 0 0 1 1 0 1 1
4 3 1 1 1 1 0 0 0 0 0 1
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```

```

1 2 0 0 0 0 0 1 1 0 1 1
4 3 1 1 1 1 0 0 0 0 0 1
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8 7 1 0 1 0 0 21 22 23 1 1
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```

```

1 2 0 0 0 0 0 1 1 0 1 1
4 3 1 1 1 1 0 0 0 0 0 1
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8 7 1 15 1 19 20 21 22 23 1 1
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```

```

1 2 0 0 0 0 0 1 1 0 1 1
4 3 1 1 1 1 0 0 0 0 0 1
5 6 0 0 0 0 1 1 1 0 1 1
8 7 1 0 1 0 0 19 20 21 1 1
9 10 1 0 0 1 1 18 1 22 1 1
0 11 12 13 14 15 16 17 1 23 24 1
0 1 1 1 1 1 1 1 1 1 25 26

```

```

1 2 0 0 0 0 0 1 1 0 1 1
4 3 1 1 1 1 0 0 0 0 0 1
5 6 0 18 19 20 1 1 1 0 1 1
8 7 1 17 1 21 22 23 24 25 1 1
9 10 1 16 15 1 1 0 1 26 1 1
0 11 12 13 14 0 0 0 1 27 28 1
0 1 1 1 1 1 1 1 1 1 29 30

```

```

1 2 0 0 0 0 0 1 1 0 1 1
4 3 1 1 1 1 0 0 0 0 0 1
5 6 0 0 0 0 1 1 1 0 1 1
8 7 1 0 1 0 0 21 22 23 1 1
9 10 1 14 15 1 1 20 1 24 1 1
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```

```

1 2 0 0 0 0 0 1 1 0 1 1
4 3 1 1 1 1 0 0 0 0 0 1
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```

```

1 2 0 0 0 0 0 1 1 0 1 1
4 3 1 1 1 1 0 0 0 0 0 1
5 6 7 8 0 0 1 1 1 0 1 1
0 0 1 9 1 0 0 17 18 19 1 1
0 0 1 10 0 1 1 16 1 20 1 1
0 0 0 11 12 13 14 15 1 21 22 1
0 1 1 1 1 1 1 1 1 1 1 23 24

```

```

1 2 0 0 0 0 0 1 1 0 1 1
4 3 1 1 1 1 0 0 0 0 0 1
5 6 7 8 0 0 1 1 1 0 1 1
0 0 1 9 1 0 0 17 18 19 1 1
0 0 1 10 11 1 1 16 1 20 1 1
0 0 0 0 12 13 14 15 1 21 22 1
0 1 1 1 1 1 1 1 1 1 1 23 24

```

```

1 2 0 0 0 0 0 1 1 0 1 1
4 3 1 1 1 1 0 0 0 0 0 1
5 6 7 8 9 10 1 1 1 0 1 1
0 0 1 0 1 11 12 13 14 15 1 1
0 0 1 0 0 1 1 0 1 16 1 1
0 0 0 0 0 0 0 0 1 17 18 1
0 1 1 1 1 1 1 1 1 1 1 19 20

```

```

1 2 3 4 5 6 7 1 1 0 1 1
0 0 1 1 1 1 8 9 10 11 0 1
0 0 0 0 0 0 1 1 1 12 1 1
0 0 1 0 1 0 0 0 0 13 1 1
0 0 1 0 0 1 1 0 1 14 1 1
0 0 0 0 0 0 0 0 1 15 16 1
0 1 1 1 1 1 1 1 1 1 17 18

```

total=180

min=18